

Traffic law violation detection using AI techniques

SECOND REVIEW REPORT

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TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	Abstract	4
1.	Introduction	5
2.	Motivation	6
3.	Problem Statement	7
4.	Literature Review	8
5.	Objective	9
6.	Methodology	10
7.	Process of Detection and Tracking of Cras taking turns	11
8.	Outputs of car taking turns	12-14
9.	Conclusion	15
10.	Future Scope	16
11.	References	17

LIST OF FIGURES

S.NO	TITLE	PAGE NO.
8.1	Turn signal information in frames	12
8.2	Turn prediction with car_IDs	13
8.3	Turn detection of car	14
8.4	Turn detection of truck	14

ABSTRACT

This report presents a comprehensive investigation into automated vehicle detection and tracking systems that harness advanced computer vision, deep learning, and signal processing techniques. Our work focuses on detecting and tracking cars especially those taking turns without proper indication.

The system leverages the YOLO model for real-time vehicle identification, combined with robust tracking via Kalman filters and temporal analysis. We incorporate feature extraction methods (such as AKAZE) and Fast Fourier Transform (FFT) analysis to optimize turn signal recognition, clearly differentiating between braking and turning actions. Our novel contributions lie in unifying spatial and temporal data to enhance accuracy under diverse lighting and environmental conditions.

Our work mainly focussed on tracing vehicles and their trajectories. We have identified the vehicles which are going straight and taking turns. Then, we collected the IDs of vehicles which are turning as well as which are turning without proper indication of turn signal.

CHAPTER 1. INTRODUCTION

Automated detection systems are critical in modern transportation as urban vehicle density increases. The emergence of computer vision and machine learning has revolutionized traffic monitoring by replacing error-prone manual processes. Our report introduces an integrated framework for vehicle detection and tracking, focusing on scenarios where cars take unindicated turns, a behavior that greatly contributes to accidents.

1.1 Prior Research Overview: Previous works in helmet detection, traffic signal violation detection, and wrong-way detection have employed methods ranging from basic image processing to advanced deep learning. These studies form the foundation upon which our system is built, yet they often operate in isolation, missing the benefits of a unified approach.

1.2 Significance of Turn Detection: Accurately detecting and tracking vehicles particularly during turns is essential for preventing collisions and ensuring efficient traffic flow. Our system captures not only the spatial features but also the temporal dynamics (e.g., hyperbolic trajectories during turns), offering real-time situational awareness.

CHAPTER 2. MOTIVATION

Improving vehicle detection and tracking is imperative given the escalating number of road accidents due to human error. Manual monitoring systems are inefficient, and current automated systems struggle with challenges such as variable lighting, occlusions, and subtle behavioral cues.

2.1 Societal Impact: Enhanced detection systems can dramatically reduce accidents by providing timely alerts to law enforcement and traffic management agencies. Reliable systems help mitigate risks associated with unindicated turns, helmet non-compliance, and wrong-way driving, thus enhancing overall road safety.

2.2 Technological Drivers: Advances in deep learning and object detection (e.g., YOLO) have created new opportunities for overcoming the limitations of existing systems. By integrating multiple detection modules into one cohesive framework, our approach promises higher accuracy and efficiency across complex traffic environments.

CHAPTER 3. PROBLEM STATEMENT

Our problem statement mainly focuses on detecting the vehicles which are turning without proper indication lights. Vehicles should start blinking when it starts moving from a stationary position, or at least 30 meters before the road intersection where turns are made. While turning left (right), the corresponding indicator lights should be blinking with a frequency of 60 to 120 times per minute.

3.1 Key Challenges:

It will be very difficult to find whether vehicle has used turn signal when intensity of turn signal is low. Also, in the morning in presence of sunlight it will be very challenging because of reflection of light from sun and surroundings.

CHAPTER 4. LITERATURE REVIEW

4.1 Helmet Violation Detection:

Early helmet detection relied on color segmentation and edge detection. More recent methods use CNNs for robust feature learning, yet challenges remain in dynamic environments.

4.2 Traffic Signal Violation Detection:

Traditional techniques employed thresholding in color spaces like HSV. Modern approaches incorporate temporal analysis and frequency-domain methods (FFT) to recognize blinking patterns in LED traffic lights, enhancing detection reliability under varying illumination.

4.3 Wrong-Way Detection Approaches:

Initial methods used simple motion tracking, but modern systems leverage deep learning and Kalman filtering to track vehicle orientation accurately. Despite progress, these methods often struggle with occlusions and non-linear trajectories.

CHAPTER 5. OBJECTIVES

Our goal is to create a unified, real-time vehicle detection and tracking system that effectively identifies traffic violations. This includes detecting vehicles taking unindicated turns, along with modules for helmet, traffic signal, and wrong-way detection.

5.1 Short-Term goal:

Implement robust YOLO-based detection, integrate Kalman filter tracking for accurate trajectory analysis, and optimize feature extraction (using AKAZE and FFT) to detect turning vehicles.

5.2 Long term goal:

Our long-term goal is to build a highly accurate and scalable traffic monitoring system that can handle large video datasets in real-world conditions. We aim to overcome challenges like low turn signal visibility and lighting interference by using advanced computer vision and machine learning techniques.

CHAPTER 6. METHODOLOGY

6.1 Detection Module:

We begin by processing video frames using the YOLO model, which generates bounding boxes around detected vehicles. Each frame is preprocessed, resized and normalized to ensure consistency.

6.2 Tracking Module:

Detected vehicles are then tracked using a Kalman filter (DEEP SORT). This step maintains continuity in vehicle trajectories over time, crucial for identifying turning maneuvers. The tracking algorithm predicts future positions and updates vehicle paths, allowing for the analysis of trajectory deviations.

6.3 Feature Extraction & Classification:

This project incorporates feature extraction and classification techniques for various traffic violation detection tasks. Specifically, turn signal detection utilizes AKAZE for visual feature extraction and Fast Fourier Transform (FFT) to analyze temporal intensity changes, enabling the system to distinguish between braking and unindicated turns. Similar deep learning and tracking methodologies are employed in additional modules for helmet detection, traffic signal violation, and wrong-way detection.

6.4 Data Logging:

The system logs detected violations with vehicle IDs, timestamps, and behavior classifications. Furthermore, photographic evidence of vehicles involved in detected violations is securely stored in cloud data.

CHAPTER 7. PROCESS OF DETECTION AND TRACKING OF CARS TAKING TURNS

Video frames are captured and preprocessed (resizing, normalization) to maintain consistency. The code reads frames sequentially, skipping alternate frames if needed for efficiency. The YOLO model processes each frame to detect vehicles, outputting bounding boxes with associated confidence scores. These boxes are then stored for tracking.

Each detected vehicle is tracked using a Kalman filter. The filter predicts future positions based on current and past locations, thereby creating continuous vehicle trajectories. The code maintains a dictionary of vehicle IDs with corresponding trajectory points, updating positions over time. For turning detection, the system analyzes trajectories. A straight-line trajectory indicates normal movement, whereas a hyperbolic or curved trajectory suggests a turning maneuver. When a deviation is detected, the system flags the vehicle for further analysis.

AKAZE extracts robust features from vehicle regions, especially around turn signals. The temporal intensity variations of these regions are processed using FFT to distinguish between braking and turning signals. The code's key functions include frame extraction, object detection, trajectory update via the Kalman filter, and feature analysis routines. Challenges such as occlusion and rapid motion are mitigated by adjusting frame intervals and tuning the Kalman filter parameters.

CHAPTER 8. OUTPUTS OF CAR TAKING TURNS

The system's outputs for tracking cars taking turns include annotated video frames where vehicles are highlighted, and trajectories are drawn. Detailed logs capture vehicle IDs, timestamps, and maneuver classifications flagged for review. Visual outputs clearly demonstrate the system's ability to differentiate between straight driving and turning behaviours through trajectory analysis.

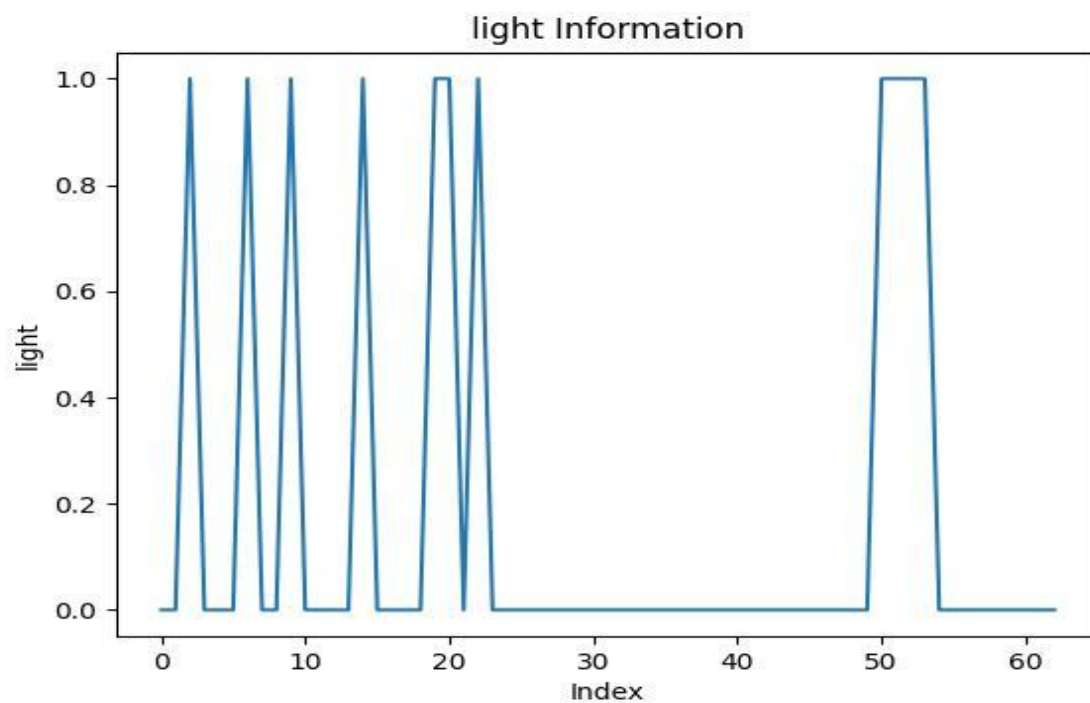


Fig 8.1: Turn signal information in frames

This graph, "light Information," visualizes the detected presence (1.0) or absence (0.0) of a vehicle's turn signal (or brake light) over frames. The x-axis represents the frame index. The fluctuating pattern between index ~1 and 22 likely indicates a blinking turn signal. The signal is mostly absent between ~23 and 50, with a brief reappearance around index 51-54. This temporal sequence of light detection is used by the code to determine if a turn signal violation occurred.

Y-axis ("light"): This axis represents the output of the light, predict function (or more specifically, the model. predict within it) for each processed frame of a specific vehicle.

Value 1,0: Likely indicates that the light detection model (light, model) detected the presence of a turn signal (or potentially a brake light, given the classes the YOLO model is trained on which often correspond to car and truck).

Value 0,0: Likely indicates that the light detection model did not detect any turn signal (or brake light) in that frame.

X-axis ("Index"): This axis represents the frame number or the index of the processed image of the vehicle within its tracked trajectory.

	A	B	C
1	video_name	car_id	predict_turn
2	intersection_surveillance	7	straight
3	intersection_surveillance	47	right
4	intersection_surveillance	1	straight
5	intersection_surveillance	37	straight
6			

Fig 8.2: Turn prediction with car IDs

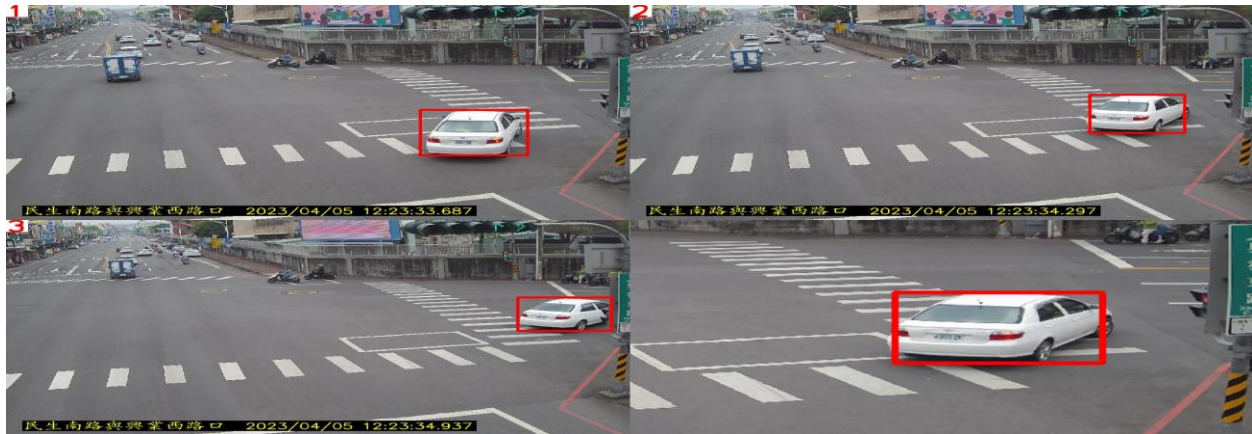


Fig 8.3: Turn detection of car

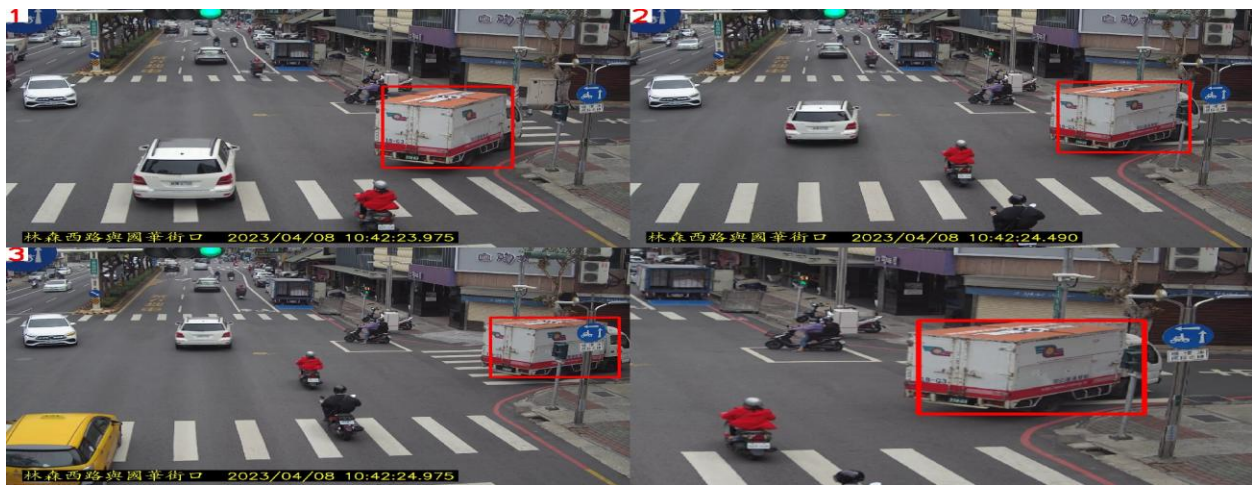


Fig 8.4: Turn detection of truck

Fig 2 is a screenshot of output in CSV file showing video name, car ID and how it has turned. Fig 3 represents violation detection of a car. Fig 4 represents turn detection of a truck.

8.1 Overall Findings: The integrated framework, combining YOLO detection, Kalman filter tracking, AKAZE feature extraction, and FFT analysis, delivers strong performance. Its key strengths include robust operation across varied environmental conditions, accurate detection of unindicated turns, minimal false positives, and real-time processing capabilities.

CHAPTER 9. CONCLUSION

This report has presented a comprehensive investigation into automated vehicle detection and tracking systems, focusing on advanced methodologies to enhance road safety. Our research demonstrates that by integrating state of the art object detection models (YOLO) with robust tracking algorithms (Kalman filters), sophisticated feature extraction techniques (AKAZE), and temporal analysis (FFT), we can reliably detect and classify vehicle maneuvers, particularly unindicated turns. The experimental results indicate strong performance across varying lighting and environmental conditions. In conclusion, our unified approach is the detection and tracking of vehicles in complex traffic environments, ensuring timely alerts and robust law enforcement support.

Implications and Future Prospects: Our findings demonstrate the potential of powerful machine learning and computer vision approaches to transform traffic surveillance. While our current approach satisfies the major objectives, there are still some hurdles to overcome, such as further lowering false positives and optimizing computing performance. Nonetheless, this research establishes a solid platform for future improvements in intelligent transportation systems. The thorough integration of geographical and temporal analysis has important implications for improving road safety and enabling proactive traffic management.

CHAPTER 10. FUTURE SCOPE

In the future, we aim to enhance our system's ability to handle large volumes of video data while maintaining high accuracy in detecting both vehicle turns and traffic violations. One of the key challenges we foresee is dealing with real-world conditions such as low turn signal visibility, interference from sunlight, and reflections from surrounding objects all of which can reduce the reliability of detection. To address these challenges, we plan to integrate a combination of advanced computer vision techniques and machine learning models that are more robust to such environmental variations.

Additionally, we intend to broaden the scope of traffic rule violation detection. This includes improving helmet detection systems, implementing littering detection, enhancing the accuracy of redlight violation detection, and refining the detection of vehicles moving in the wrong direction on one-way roads. The goal is to build a scalable, adaptable system that can be applied to a wide range of urban traffic monitoring scenarios.

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